



Amir Adam

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ABOUT ME

Embedded Systems and IoT engineering student with hands-on experience in firmware development using C/C++ and Python on ESP32, STM32, and Raspberry Pi. Strong in communication protocols, system integration, and building end-to-end solutions for sensor interfacing, data acquisition, and network communication. Seeking to apply practical problem-solving and debugging skills in real-world engineering environments.

EDUCATION AND TRAINING

16/08/2022 – CURRENT Mohali, India

BACHELOR OF ENGINEERING IN ELECTRONICS & COMMUNICATIONS Chandigarh University

Relevant Courses: Embedded System, VLSI, Communication

Level in EQF EQF level 6

13/02/2017 – 01/12/2020 Wau, South Sudan

CERTIFICATE OF SECONDARY EDUCATION St. Ignatius of Loyola Secondary School

General Sciences: Maths, Physics, Chemistry, Biology

Level in EQF EQF level 4

SKILLS

Programming

C/C++ | Python | Typescript

Embedded Systems

Microcontrollers: ESP32, STM32 | Raspberry Pi | Sensor Integration

Tools & Platforms

Git | STM32CUBEIDE | Arduino IDE | VSCode | GitHub

App Development

React Native | Supabase | Expo/Expo router | PostgreSQL

Conceptual Knowledge

Communication Protocols: I2C, SPI, UART | Embedded Systems Architecture | Networking Basics (HTTP, client-server side) | Basic PCB Design

LANGUAGE SKILLS

Mother tongue(s): **ARABIC(SUDANESE DIALECT)**

Other language(s):	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
	ENGLISH	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **PROJECTS**

10/03/2026 – 24/03/2026

Edge AI Surveillance System with Real-Time Intrusion Detection (Raspberry Pi 5)

- Technologies:** Python , OpenCV ,TensorFlow Lite , SQLite, FastAPI, Telegram Bot API
- Designed and implemented a real-time edge AI surveillance system on Raspberry Pi 5 for human detection using a TensorFlow Lite MobileNet SSD model.
 - Built a modular processing pipeline including video ingestion, preprocessing (OpenCV), AI inference, and spatial decision logic for intrusion detection.
 - Developed a custom spatial logic engine to evaluate object position (center of mass and foot approximation) against defined restricted zones.
 - Implemented asynchronous alerting using multithreading, delivering real-time intrusion notifications (image + metadata) via Telegram Bot API.
 - Engineered a local web interface using FastAPI to stream annotated MJPEG video and visualize detection logs stored in SQLite.
 - Integrated stateful logging for event tracking, including timestamps, confidence scores, and snapshot storage for post-event analysis.

Link <https://github.com/amirphiladam2/edge-ai-surveillance>

15/02/2026 – 19/02/2026

Full-Stack IoT Monitoring System with Web Dashboard

- Technologies:** ESP32, React(Vite), Python (Flask), HTTP, REST API
- Designed and developed a full-stack IoT system for real-time sensor data acquisition and visualization.
 - Implemented firmware on ESP32 to collect environmental sensor data and transmit it to a backend server via HTTP requests.
 - Built a RESTful API using Flask to handle data ingestion, processing, and storage.
 - Developed a responsive frontend dashboard using React.js to display real-time and historical sensor data.
 - Enabled dynamic data updates and visualization through API integration between frontend and backend.
 - Improved system reliability by handling network delays and ensuring consistent data transmission.

Link <https://github.com/amirphiladam2/FullStack-IoT>

20/10/2025 – 20/11/2025

Disaster-Resilient Offline Mesh Communication System

- Technologies:** ESP32, LoRa
- Designed and developed an offline mesh communication system using ESP32 microcontrollers and LoRa to enable connectivity in network-isolated environments.
 - Implemented peer-to-peer mesh networking architecture to support decentralized message transmission without reliance on cellular or internet infrastructure.
 - Integrated sensor data acquisition and local processing to enhance situational awareness in emergency scenarios.
 - Optimized embedded firmware for low-power consumption, improving system reliability in constrained and disaster-prone environments.

Link <https://techtacklenerd.blogspot.com/2025/11/disaster-resilient-hybrid-mesh.html>

20/10/2024 – 22/10/2024

Home Automation System using ESP32 and PySerial

- Technologies:** ESP32, Python (PySerial)
- Designed a serial communication-based home automation system for controlling electrical appliances.
 - Programmed ESP32 firmware to receive and execute control commands via UART communication.

- Integrated relay modules to enable switching of AC loads (lamps) based on microcontroller signals.
- Developed a Python-based interface using PySerial to transmit real-time control commands from a computer to ESP32.

Link <https://techtacklenerd.blogspot.com/2024/10/home-automation-project-using-esp32-and.html>

CREATIVE WORKS

12/07/2023 – CURRENT

YouTube Tech Content Creator

- Created technical educational content focused on Embedded Systems and App development concepts.
- Produced video tutorials demonstrating real-world implementations using STM32,ESP32, React Native.

Link <https://www.youtube.com/@amirdevstudios/videos>

HOBBIES AND INTERESTS

Hobbies

- Nature photography, with a focus on landscapes and environmental elements
- Coding and Electronics Tinkering